ASTRUM DRIVE

# ASTRA

## Compact, Verifiable AI for Scientific Discovery

Astrum Drive's internal multi-model research capability, turning scientific objectives into falsifiable conjectures, reviewed executable tests, reproducible evidence and the next decision.

### Company Capability Brief for Investors, Clients and Strategic Partners

Strategic role, current benchmark evidence, competitive reference architecture, algorithmic workflow, operating economics, capability roadmap and explicit limits.

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# Executive thesis

## ASTRA makes AI-generated scientific reasoning inspectable.

Most AI systems optimize the answer. ASTRA optimizes the evidence chain: independent models propose and challenge a claim, a specialist writes a falsifiable validator, another model audits that validator, computation engines execute it, and a human controls what becomes accepted knowledge.

## The problem

Scientific and engineering teams can generate hypotheses and code faster with AI, but the bottleneck moves downstream: deciding whether the produced result actually tests the intended claim. Plausible prose, a successful program, or a printed PASS is not sufficient evidence.

This creates four practical risks:

- hidden changes in assumptions, domains or boundary conditions;
- validators that confirm their own construction;
- numerical checks presented as universal proofs;
- fragmented evidence across chats, notebooks and tools.

## The ASTRA response

ASTRA coordinates complementary AI models and deterministic scientific engines around one shared objective. Generation, code authorship, code review, execution, result analysis and research navigation are deliberately separated.

The system is designed to produce:

- a precise and falsifiable statement;
- executable validation code with explicit failure paths;
- raw output from symbolic, numerical or logical engines;
- a conservative interpretation tied to the original objective;
- a persistent record that experts can reproduce and challenge.

**Company asset proposition.** ASTRA is Astrum Drive's internal orchestration and assurance layer for AI-assisted R&D. It is model-flexible, engine-extensible and deployable close to private data and scientific software. ASTRA is not currently offered for sale, subscription or license. Its economic role is to strengthen the speed, quality, traceability and capital efficiency of the research, engineering and client work performed by Astrum Drive.

**Category reference.** Google DeepMind's Deep Think, Aletheia, Co-Scientist and AlphaEvolve demonstrate that long-horizon search, debate, verification and automated evaluation can extend frontier models in science. ASTRA adopts that general direction but makes a different architectural choice: heterogeneous models, reviewed executable validators, local or private compute, and evidence bundles that can be inspected by Astrum Drive, collaborators and authorized project stakeholders.

## What exists today

| 68/68 | 8/8 | 100% |
|---|---|---|
| automated production tests passing | local–ASTRUM evidence bundles passing | cross-oracle agreement on four paired cases |

The current system includes a browser application, an MCP interface for agent control, local and remote execution, detached long-running jobs, benchmark suites, report persistence, formal proof and CAS routes, package extensions, bounded repair and human approval gates. Validator Repair vNext.1 compiles and audits code before model review, applies safe source-local fixes in milliseconds and restricts Claude to one exact, auditable patch rather than a wholesale regeneration.

**Evidence position, not a leaderboard claim.** ASTRA currently demonstrates operational reliability, cross-environment reproducibility and useful multi-model controls. In a four-case paired public-benchmark canary, the heterogeneous proposal team produced greater measured perspective separation than the homogeneous control (0.7459 versus 0.5958), but both passed 2/4 cases. It does not yet demonstrate higher scientific success than Deep Think, Aletheia or other frontier systems. The frozen long-horizon canary is the release gate for that stronger claim; the first vNext.0 result was a valid null result.

# Part I – Strategic company capability

## One internal capability, three beneficiaries

| Beneficiary | Immediate value | Typical application |
|---|---|---|
| **Astrum Drive R&D** | Move from a research question to a reproducible computational test while preserving assumptions and failed paths. | Theoretical physics, symbolic models, differential equations, optimization, simulation validation. |
| **Astrum Drive clients** | Receive stronger analyses, engineering decisions and evidence packages produced by Astrum Drive with ASTRA support. | Model verification, solver regression, design-space exploration, mathematical and scientific QA. |
| **Investors and partners** | Gain exposure to a controlled scientific capability that compounds across the company's projects and proprietary methods. | Company financing, joint research, compute, model and scientific-software integrations. |

## The company job

**When** a technical team has a consequential scientific claim, **it needs** to translate that claim into a test that can fail, execute it with the right tools, and preserve enough evidence for another expert to reproduce the decision. **ASTRA gives Astrum Drive that governed research cycle as a reusable internal capability.**

## Candidate use cases

- symbolic identities, invariants and limiting cases;
- general relativity metrics and curvature calculations;
- quantum operators, states and consistency checks;
- differential equations and boundary-value problems;
- fluid mechanics and dimensional validation;
- counterexample search with Z3;
- formal logic and Lean 4 proofs;
- optimization using internal scientific packages;
- cross-validation through Mathematica;
- reproducibility audits for model-generated code.

## Value creation

### Faster path to evidence
Automates the transitions between hypothesis, formalization, executable test, correction and interpretation.

### Lower verification friction
Places a code review and deterministic execution between model output and scientific acceptance.

### Reusable organizational knowledge
Stores validated results, refutations, scripts and reports instead of losing them in isolated conversations.

### Flexible scientific stack
Routes work through local, remote, open-source, formal and in-house computation engines.

### How ASTRA creates economic value

ASTRA is not a standalone commercial offering. Its value reaches the market through Astrum Drive's work and through the capabilities it helps the company build:

- faster and more reproducible internal physics and engineering research;
- higher-value client analyses, designs, validation programs and evidence bundles;
- reuse of verified methods across projects instead of rebuilding each workflow;
- stronger due diligence, partner confidence and defensible technical decisions;
- integration of proprietary packages, compute and accumulated scientific know-how.

**Ownership boundary.** Clients may purchase Astrum Drive research, engineering, validation or collaborative project outcomes. ASTRA itself remains an internal company resource unless Astrum Drive deliberately changes that strategy in the future.

## Why ASTRA can be differentiated

| | Conventional AI assistant | ASTRA |
|---|---|---|
| Primary output | A response or code artifact | A traceable evidence chain |
| Model structure | Often one model and one context | Multiple models with separated responsibilities |
| Validation | Optional, user-initiated | Mandatory execution in the core cycle |
| Code quality control | Same model may author and judge | Independent reviewer can reject and request revision |
| Scientific engines | Ad hoc tool use | Routed oracle with explicit engine metadata |
| Failure semantics | Errors may be folded into prose | `REFUTED`, `CODE_ERROR`, `API_ERROR`, weak evidence |
| Knowledge governance | Chat history | Reports, branches, axiomatic base and human gate |
| Deployment posture | Primarily hosted | Local-first with optional remote execution |

### Defensibility thesis

**Workflow IP and calibration.** The valuable asset is not a single prompt. It is the evolving set of role contracts, review rules, verdict guards, correction behavior, benchmark cases and domain-specific validation patterns.

**Engine and package graph.** ASTRA can accumulate verified adapters and test protocols across symbolic algebra, SMT, numerical science, formal proof, Mathematica and proprietary packages.

**Evidence and failure data.** With appropriate project consent and governance, recurring validator defects, refutation patterns and correction outcomes can improve system calibration.

**Model independence.** The orchestration layer can adopt stronger models without rebuilding the scientific workflow. Models are replaceable capabilities; the evidence architecture persists.

## Reference landscape: frontier research agents

The category is moving from one-shot answers toward compound systems that search, debate, verify and evolve candidate work. Google DeepMind reports professional mathematics and science results from Gemini Deep Think and its Aletheia research agent.[1] Aletheia iteratively generates, verifies and revises natural-language solutions.[2] Google's Co-Scientist uses a multi-agent generate–debate–evolve process for

[1] Google DeepMind, "Accelerating Mathematical and Scientific Discovery with Gemini Deep Think," 2026.
[2] Feng et al., "Towards Autonomous Mathematics Research," arXiv:2602.10177, 2026.

hypothesis generation,[3] while AlphaEvolve combines a fast and a deep model with automated evaluators and evolutionary search over programs.[4]

| System | Publicly described strength | Relationship to ASTRA |
|---|---|---|
| **Gemini Deep Think / Aletheia** | Frontier reasoning plus a long-horizon generator–verifier–reviser loop for professional mathematics and scientific problems. | Closest epistemic reference. ASTRA does not claim comparable published research performance; it externalizes verification into reviewed code and Astrum-controlled or project-authorized engines. |
| **Google Co-Scientist** | Asynchronous specialized agents generate, debate, rank and evolve research hypotheses with scientist guidance. | Validates multi-agent scientific collaboration as a category. ASTRA uses a smaller fixed role map and emphasizes executable evidence after hypothesis generation. |
| **AlphaEvolve** | An ensemble balances breadth and depth while automated evaluators drive evolutionary program search. | Shares evaluator-grounded iteration. ASTRA targets heterogeneous scientific workflows rather than domains with one cheap scalar evaluator. |
| **ASTRA** | Codex, Antigravity and Claude separate conjecture, navigation, implementation and review; deterministic oracles decide executable checks. | Local-first assurance layer for physics, mathematics and engineering, with package adapters, human approval and auditable failure semantics. |

## The shared principle and the distinct architectural bet

**Shared principle.** Verification is a process rather than a final prompt. Candidate work must be challenged, revised or discarded; proof and refutation should be pursued together; tools can extend reasoning; and an expert remains responsible for consequential acceptance.

**ASTRA's architectural bet.** Astrum Drive needs model plurality, validator-code review, engine-grounded evidence, private execution, internal-package integration and an organizational approval trail. ASTRA assembles those controls without requiring Astrum Drive to train a new foundation model.

**No false equivalence.** Deep Think, Aletheia, Co-Scientist and AlphaEvolve are frontier research programs with reported results that ASTRA has not reproduced. Their public disclosures do not provide an apples-to-apples per-discovery cost. ASTRA's current claim is narrower: useful controls and reproducible evidence can be delivered with a compact three-model orchestrator and a single scientific workstation.

[3] Google DeepMind, "Co-Scientist: A multi-agent AI partner to accelerate research," 2026.
[4] Google DeepMind, "AlphaEvolve," 2025.

# Benchmark evidence as of July 26, 2026

## What has been measured

| Evidence gate | Result | Interpretation |
|---|---|---|
| **Automated production suite** | 68/68 passed | Core routing, deliberation, Lean safeguards, external evaluators, trajectory metrics, validator repair and application evidence remain regression-tested. |
| **Local application validation** | 5/5 passed | Z3, SciPy, Pint, SymPy and GR_python produced complete evidence bundles in the production Python environment. |
| **Local–ASTRUM replication** | 8/8 passed; 100% agreement | Four application cases produced matching claim verdicts on Windows and the remote Linux workstation. This is reproducibility evidence, not broad scientific accuracy. |
| **Parallel execution smoke** | 8/8 grade-A runs | Four deterministic cases ran on both oracles with four workers: verdict accuracy and cross-oracle agreement were both 100%; P50/P95 latency was 5.422/8.598 seconds. |
| **vNext.1 local repair gate** | 4/4 repaired | Every exact source pattern that the deterministic repairer claims to support was detected and repaired; median/P95 repair time was 0.702/1.913 ms. |
| **Sound-code calibration** | 0/3 false blocks | Three sound validators, including an EinsteinPy API regression, remained approved by the deterministic gate. The sample is intentionally small and must grow. |
| **Paired diversity canary** | 2/4 passed in both arms | Heterogeneous proposers increased mean perspective separation from 0.5958 to 0.7459, but produced no pass-rate uplift in four public cases. |
| **Long-horizon vNext.0 canary** | 0/4 credible cycles | The first repair strategy did not improve conversion and averaged 1,045 seconds per cycle. ASTRA records this as a null result, not a success. |
| **vNext.1 one-cell diagnostic** | reviewer timeout; repaired artifact 12/12 | The maximum-model cycle stopped at the 240-second reviewer ceiling after 1,451 seconds. A diagnostic ASTRUM run exposed one noncanonical symbolic-zero failure; the new deterministic repair made all twelve checks pass. This remains unreviewed diagnostic evidence, not an admitted scientific result. |

**Why the null results are useful.** vNext.0 showed that asking Claude to regenerate large validators more often increased latency without clearing the reviewer. vNext.1 therefore applies safe deterministic edits first, performs compile/import smoke checks, allows at most one bounded exact-edit model patch, and refuses scripts too large for that patch context. The one-cell vNext.1 diagnostic then added a third local repair: canonicalization of raw symbolic tensor-zero checks. This is a measured improvement in control and cost exposure; the full comparative canary must still determine whether it improves scientific conversion.

## External benchmark strategy

ASTRA has pinned adapters or protocols for SciCode, miniF2F, FrontierScience and AInsteinBench. SciCode evaluates code generation on scientist-curated research problems,[5] FrontierScience tests expert-level scientific reasoning,[6] and AInsteinBench targets coding agents inside scientific software repositories.[7] AstaBench is a useful future umbrella because it spans literature, data, planning, tools, coding and search across more than 2,400 problems.[8]

[5] Tian et al., "SciCode: A Research Coding Benchmark Curated by Scientists," 2024.
[6] OpenAI, "Evaluating AI's ability to perform scientific research tasks," 2025.
[7] ByteDance Seed, "AInsteinBench," accessed July 2026.
[8] Allen Institute for AI, "AstaBench," accessed July 2026.

**Benchmark discipline.** Adapter readiness is not a score. Operational errors are excluded from scientific pass rates, architecture comparisons use frozen cases and budgets, and qualitative depth or novelty requires architecture-blinded expert review. The machine-readable release snapshot is tracked with the source repository.

## Compact operating profile

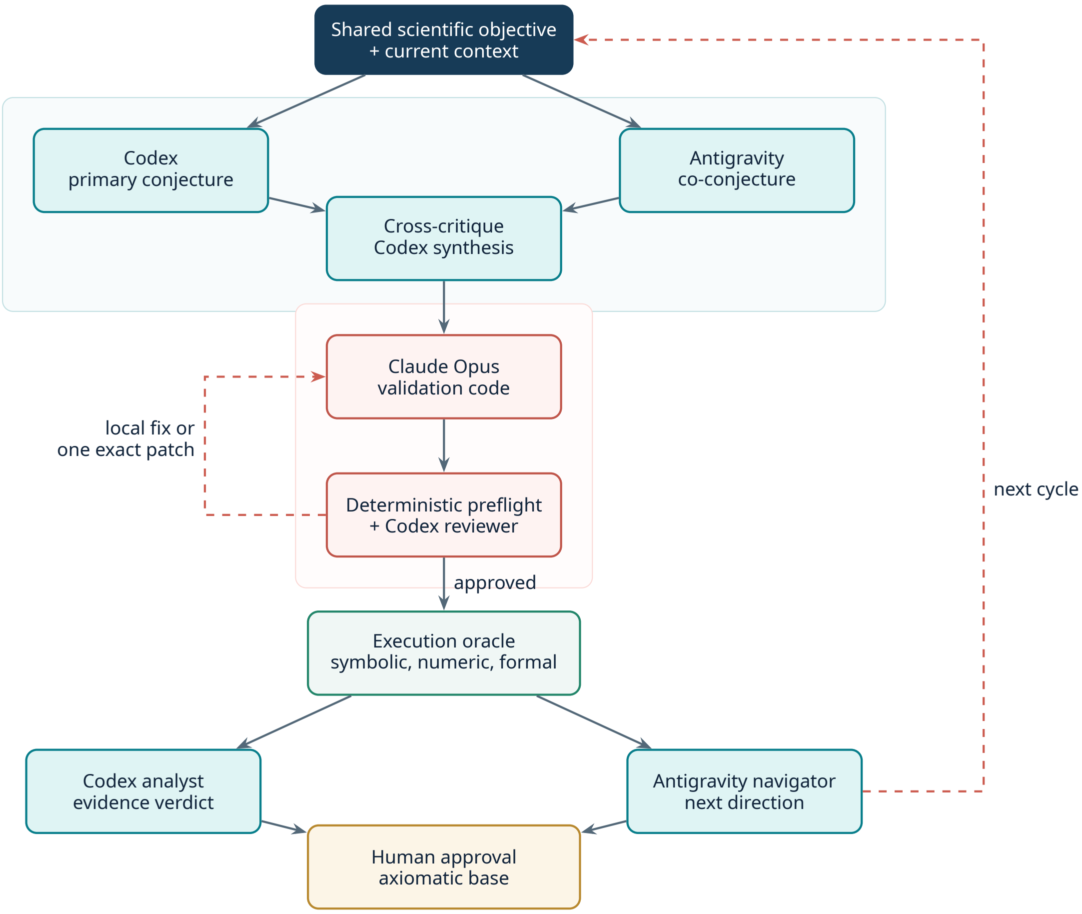
| Resource dimension | ASTRA production profile |
|---|---|
| **Foundation-model development** | No model pretraining or fine-tuning is required; ASTRA orchestrates authenticated model services. |
| **Model portfolio** | Codex `gpt-5.6-sol` at `xhigh`, Antigravity/agy `gemini-3.1-pro-high`, and Claude Code `claude-opus-4-8`, with phase-specific roles and recorded bounded fallbacks. |
| **Scientific compute** | Local workstation plus one 16-core / 32-thread ASTRUM node with 29 GiB RAM and an RTX 3080; heavy validators route remotely. |
| **Validation software** | Primarily open scientific engines plus optional Mathematica and internal packages. |
| **Repair exposure** | Deterministic edits first; at most one bounded model patch in vNext.1. |
| **Measured cost proxies** | Model-call count, phase wall time, oracle time, repair attempts and evidence yield per call. |
| **Unmeasured economics** | No audited total-cost-per-accepted-result comparison with frontier systems yet. |

## Metrics for internal and client-project value

| Operating metric | Why it matters |
|---|---|
| Time and model calls to first reproducible test | Measures research-cycle compression and inference exposure |
| Independent reviewer rejection and recovery | Measures value of heterogeneous perspectives |
| False-acceptance rate on adversarial holdouts | Core assurance and trust metric |
| Cross-oracle reproduction rate | Tests portability and environmental independence |
| Human review time per accepted result | Tests whether traceability reduces expert burden |
| Accepted evidence per dollar or model-access hour | Required before making a cost-superiority claim |

# Part II – System and algorithmic detail

## Production architecture




**Role separation and bounded correction are the central controls.** Codex leads conjecture, synthesis, code review and evidence analysis. Antigravity contributes an independent conjecture and maintains research direction. Claude authors validation code and may supply one exact-edit repair. ASTRA first applies safe deterministic fixes and compile/import smoke checks. Deterministic engines, not model consensus, produce the evidence.

## Interfaces and deployment

| Layer | Current implementation |
|---|---|
| **Interaction** | Browser interface, command-line subprocess API and MCP tools for agents |
| **Orchestration** | Asynchronous Python pipeline with progress, timeouts, fallbacks and detached jobs |
| **Models** | Subscription-authenticated Codex, Claude Code and Antigravity CLIs; optional API providers |
| **Execution** | Local Python and optional remote Linux worker; engine-specific routing |
| **Persistence** | Reports, cycle state, research branches, benchmark results and human approval |
| **Security posture** | Credentials remain in local configuration; private execution is supported; operational controls remain a hardening priority |

### Fast and deep execution profiles

**What can run in parallel.** Independent validation cases, random seeds, parameter sweeps and cross-oracle replications can fan out across local workers and ASTRUM. Within a single scientific cycle, model roles remain ordered where one artifact depends on another. Subscription-backed model calls are kept sequential in comparative benchmarks so contention and quota timing do not become hidden confounders.

ASTRUM exposes 16 physical / 32 logical CPU threads, 29 GiB RAM and an RTX 3080. CPU-parallel SciPy, NumPy and symbolic workloads use the node when appropriate; GPU work routes only when the validator is genuinely GPU-shaped. The fast release profile uses deterministic regression and replay gates first. The deep profile runs frozen, long-horizon model trajectories and architecture-blinded expert scoring.

In the current four-worker smoke, eight deterministic local/ASTRUM runs all earned grade A, with 100% verdict accuracy and cross-oracle agreement. This demonstrates parallel execution reliability and portability; it does not measure multi-model scientific depth.

## The core algorithm

**Inputs:** shared objective $G$, established context $K$, current direction $D$, maximum code revisions $R$, execution policy $O$.

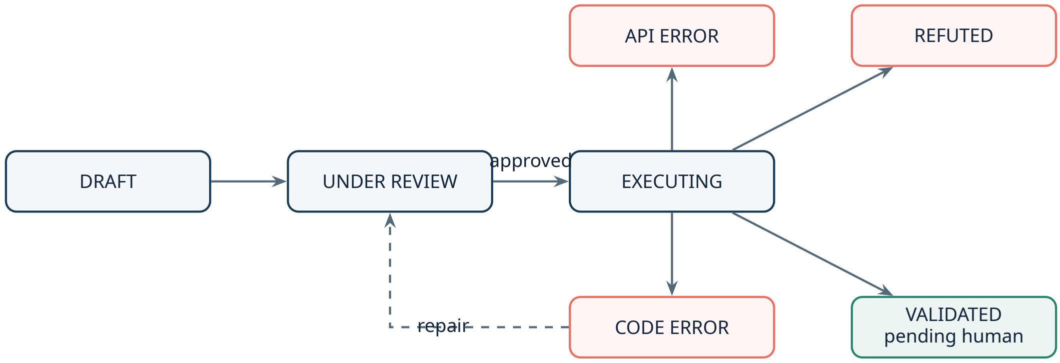
1. $p_c \leftarrow$ Codex.propose$(G, K, D)$
2. $p_a \leftarrow$ Antigravity.propose$(G, K, D)$ `[parallel]`
3. $q_c \leftarrow$ Codex.critique$(p_a)$; $q_a \leftarrow$ Antigravity.critique$(p_c)$
4. $h \leftarrow$ Codex.synthesize$(G, p_c, p_a, q_c, q_a)$
5. $v_0 \leftarrow$ Claude.translate$(G, h)$
6. $(d, c) \leftarrow$ Preflight.audit_and_compile$(v_0)$
7. $v_1 \leftarrow$ LocalRepair$(v_0, d)$ `[safe patterns only]`
8. $r \leftarrow$ Codex.review$(G, h, v_1, c)$
9. `if` $r =$ REVISE `and patchable:`
   $p \leftarrow$ Claude.exactPatch$(v_1, r)$; $v_2 \leftarrow$ ApplyBounded$(v_1, p)$
10. $e \leftarrow$ Oracle.execute$(v_2, O)$
11. $s \leftarrow$ Guard$(v_2, e)$
12. $a \leftarrow$ Codex.analyze$(G, h, v_2, e, s)$
13. $n \leftarrow$ Antigravity.navigate$(G, K, h, a)$
14. HumanGate$(h, v_2, e, a, n)$

### Conservative decision semantics

The orchestration enforces several monotonic constraints:

- a crashed validator cannot become `VALIDATED`;
- an explicit `FAIL` cannot be promoted by explanatory prose;
- a printed PASS is evidence, not authority;
- missing optional engines fail explicitly;
- a reviewer can reject circular, unreachable or sampling-only proof logic;
- final admission to the axiomatic base requires a human decision.

### State model




**Interpretation boundary.** In ASTRA, `VALIDATED` means supported by the recorded validator, execution and assumptions. It does not mean absolute truth. The evidence remains open to stronger proof, independent replication and expert review.

## Scientific engine and extension model

### Oracle portfolio

| Engine class | Examples | Validation role |
|---|---|---|
| **Symbolic** | SymPy, SageMath, Maxima | Exact simplification, identities, derivatives, algebraic residuals |
| **Logic / proof** | Z3, Lean 4 + Mathlib | Counterexample search, satisfiability, kernel-checked theorems |
| **Numerical** | NumPy, SciPy, mpmath, QuTiP | ODEs, high-precision checks, quantum systems, sweeps |
| **Domain packages** | EinsteinPy, fluids, Pint | Curvature, fluid models, units and physical consistency |
| **Independent CAS** | Mathematica bridge, Cadabra | Cross-engine validation, tensor algebra, notebook workflows |
| **Internal software** | GR_python, pyWarpFactory, warp-bubble optimization | Reuse of proprietary or project-specific research capabilities |

### Extension contract

Any scientific package can become part of the oracle when a validator:

1. declares reproducible inputs, versions and assumptions;
2. imports the package in the selected environment;
3. computes a quantity that discriminates the claim from its negation;
4. exposes exact output, errors and tolerances;
5. includes reachable failure behavior;
6. preserves the script and result for review.

### Agent control through MCP

ASTRA exposes operations for direct execution, complete cycles, progress probes, detached jobs, job polling and remote-worker status. A conversational agent can therefore plan a test and invoke ASTRA without manually copying code, while ASTRA retains the evidence and decision semantics.

**Illustrative operator instruction:**
"Formulate two competing explanations for this curvature result. Use ASTRA to construct a validator with an exact symbolic leg and an independent numerical leg. Do not accept a printed PASS until the code, residuals and assumptions have been reviewed."

### Operational hardening priorities

- workspace authentication and role-based access;
- signed evidence bundles and immutable audit metadata;
- dependency and container pinning;
- job isolation and compute quotas;
- private model and on-premise connectors;
- team review queues and approval policy;
- observability, cost and latency dashboards;
- benchmark governance and release qualification.

## Capability development roadmap

| Phase | Capability objective | Evidence required before advancing |
|---|---|---|
| **Now** | Operational multi-model research loop, vNext.1 assurance and scientific oracle | Maintain 100% release-suite pass; harden maximum-model phase budgets; complete the frozen comparative canary; publish evidence snapshots |
| **Internal scale** | Use ASTRA repeatedly across Astrum Drive research programs | Measured time-to-evidence, reviewer correction value, reproduction rate and expert satisfaction |
| **Project integration** | Governed approvals, scalable jobs and evidence delivery inside selected client or collaborative projects | Repeat use, security acceptance and measurable contribution to project quality, cycle time or margin |
| **Domain depth** | Domain-specific validators, packages and benchmark suites | Comparable or better task-level utility in selected high-value workflows |
| **Infrastructure resilience** | Private execution, partner engines and replaceable model routes | Reliable operations, controlled data paths and sustainable operating economics |

### Near-term investment and partnership priorities

1. **Capability engineering:** packaging, secure execution, observability, collaboration and evidence management.
2. **Scientific calibration:** adversarial benchmarks, false-acceptance measurement and expert-reviewed domain suites.
3. **Application programs:** two or three internal or collaborative workflows with high verification cost and measurable research-cycle value.
4. **Infrastructure partnerships:** model, compute, scientific-software and research-institution integrations that strengthen Astrum Drive's internal capability.

### Risks and mitigations

| Risk | Why it matters | Mitigation direction |
|---|---|---|
| **Validator mismatch** | Code may test a weaker or different claim | Independent code review, human gate, benchmarked failure patterns |
| **Model latency and quota** | Maximum-quality cycles can take tens of minutes | Deterministic repair, one bounded patch, phase budgets, caching, detached runs and ASTRUM fan-out |
| **False confidence** | Users may overread `VALIDATED` | Explicit semantics, evidence display, independent engines, approval policy |
| **Scientific breadth** | No system is expert in every domain | Vertical focus, package adapters, domain reviewers and scoped claims |
| **Operational security** | Generated code executes with real resources | Isolation, allowlists, containers, credentials policy and audit logs |
| **Value attribution** | ASTRA's contribution can disappear inside a broad research program | Instrument selected workflows and compare cycle time, evidence quality, expert effort and cost against a frozen baseline |

### The next proof points

- rerun the maximum-model one-cell diagnostic after phase-budget hardening, then complete the frozen vNext.1 comparison against baseline and vNext.0;
- expand adversarial holdouts beyond the four deterministic repair patterns;
- publish scored SciCode, FrontierScience, miniF2F and AInsteinBench subsets with operational failures reported separately;
- quantify expert time, model calls and compute cost per accepted or refuted claim;
- demonstrate ASTRA's measured contribution in one recurring Astrum Drive research or client-project workflow.

## Engaging with Astrum Drive

### The company capability thesis in one sentence

ASTRA is Astrum Drive's internal verification and coordination layer for using frontier AI in scientific work while preserving falsifiability, reproducibility, human control and lower capital intensity.

### For a prospective client

Engage Astrum Drive on one consequential workflow with:

- a clear scientific decision and current baseline process;
- existing code, equations, simulations or internal packages;
- measurable verification effort or rework;
- an expert who can judge whether ASTRA preserves the real claim.

Astrum Drive may use ASTRA internally to execute and validate the work. The engagement delivers the agreed research, engineering or validation outcome and its evidence; it does not sell or license ASTRA. A project should compare baseline and ASTRA-assisted work on time-to-evidence, reproducibility, review effort and false acceptance.

### For an investor or VC

The immediate diligence questions are concrete:

1. Does heterogeneous review materially improve generated validators under equal budgets?
2. In which Astrum Drive programs does ASTRA create the greatest measurable lift?
3. Can the evidence architecture become a repeatable company workflow?
4. Which data, adapters and calibration assets compound with usage?
5. Can controlled execution and scientific specialization improve project economics despite frontier-model dependence?

## For a strategic partner

Partners can strengthen Astrum Drive's governed research workflow through:

- frontier-model access for credible scientific demonstrations;
- compute infrastructure for traceable high-value workloads;
- scientific-software and agent-native interfaces;
- joint research programs that require reproducible AI-assisted methods.

ASTRA remains Astrum Drive's internal resource within these relationships; the collaboration concerns research objectives, infrastructure or outcomes rather than the sale of the orchestration system.

**Proposed next step.** Define a 4–8 week collaborative study around one scientific workflow, one baseline, one reproducibility target and an agreed set of acceptance metrics. The goal is not a demo; it is evidence that ASTRA changes a real Astrum Drive R&D or client-project decision.

**Astrum Drive value proposition.** ASTRA turns frontier-model capability into a governed scientific workflow that Astrum Drive can own, calibrate and integrate. The asset is not a claim to have built a larger model, nor a standalone software product offered for sale. It is the internal control plane, validator library, benchmark discipline, package graph and accumulated evidence that make Astrum Drive's physics and engineering work more capable, reproducible and capital-efficient.

# ASTRA

From plausible answers to reviewable evidence.

Nelson Bolívar | Founder and Lead Architect | Astrum Drive

Company capability discussions | Investors, clients and strategic partners | July 2026